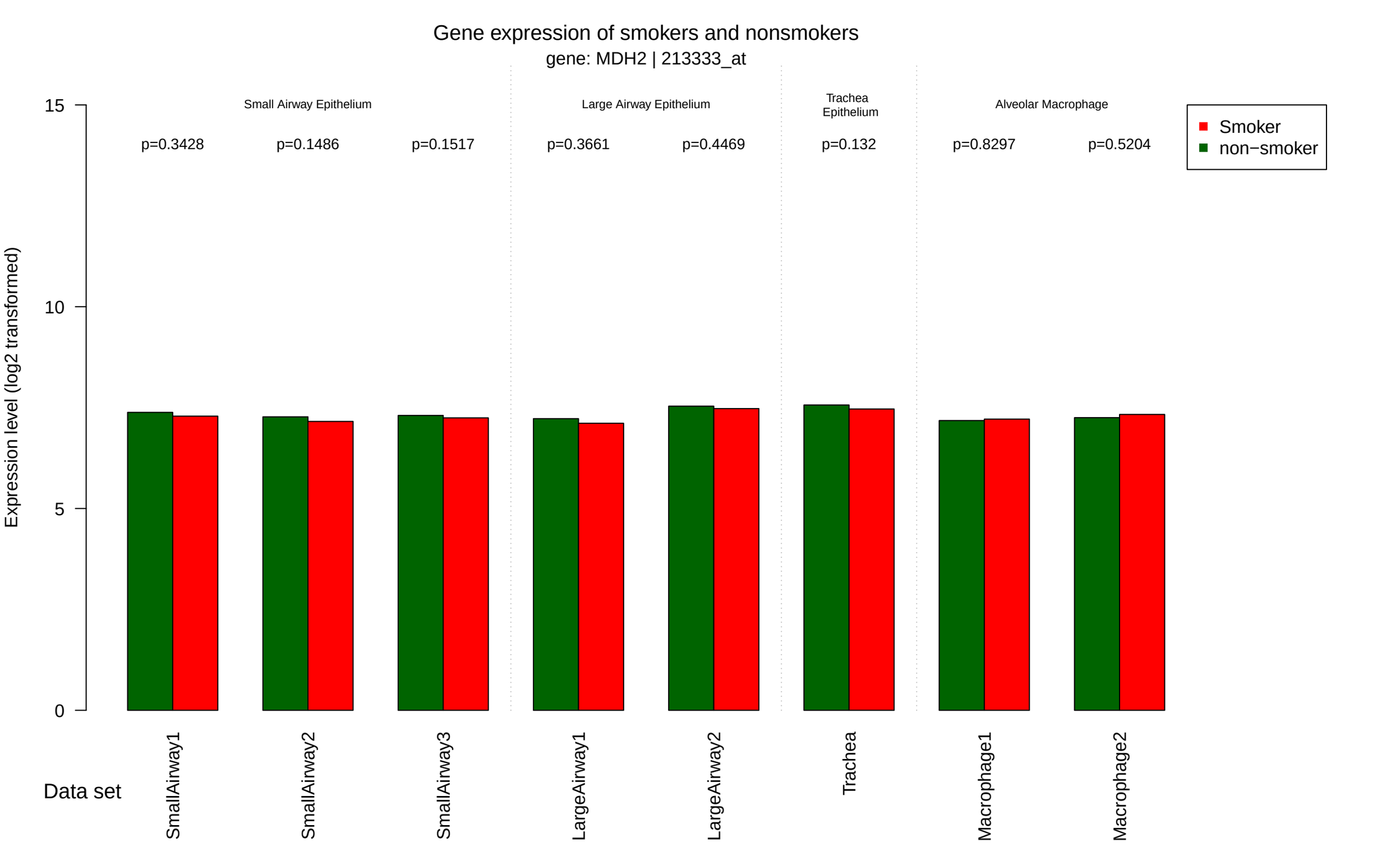
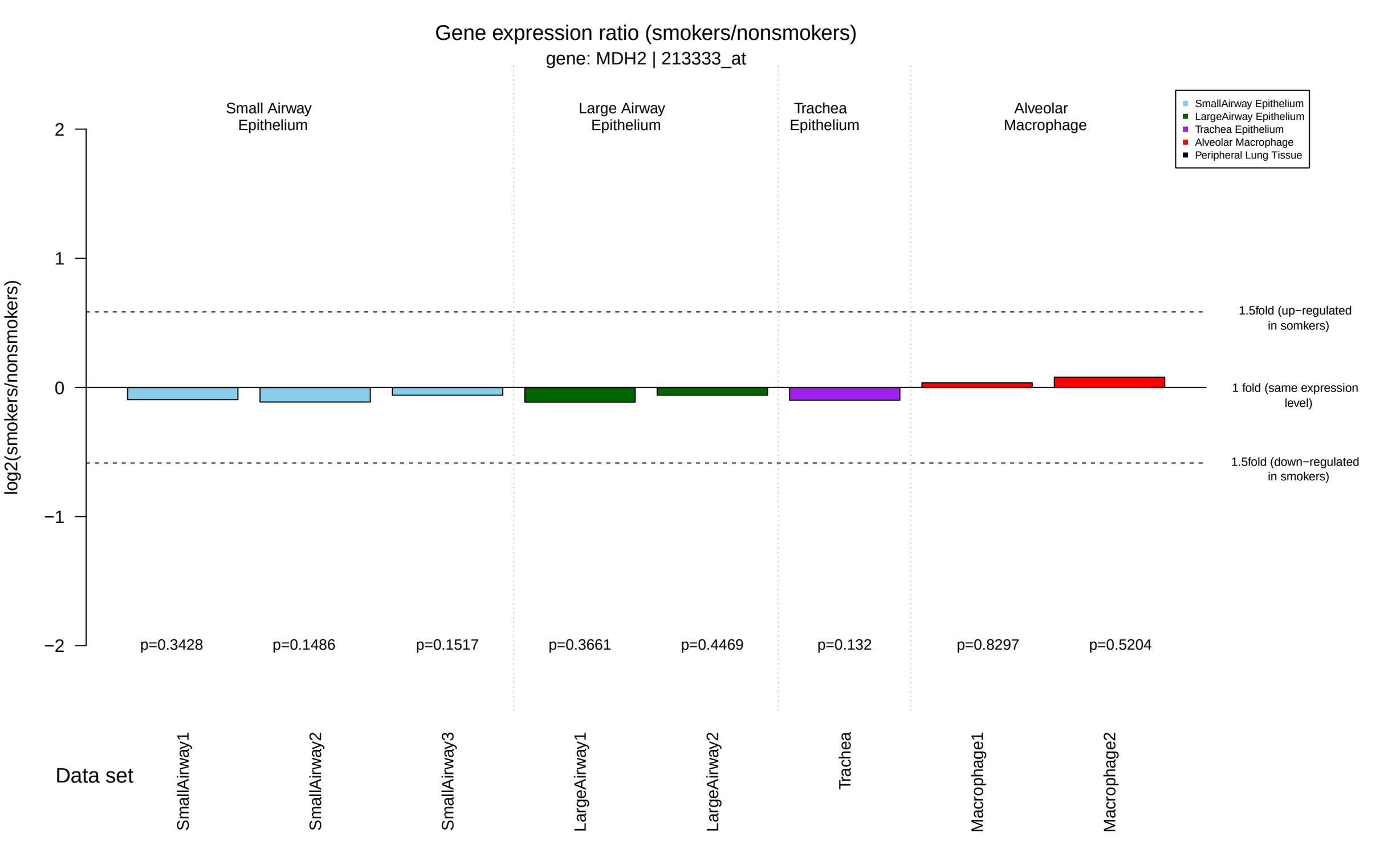
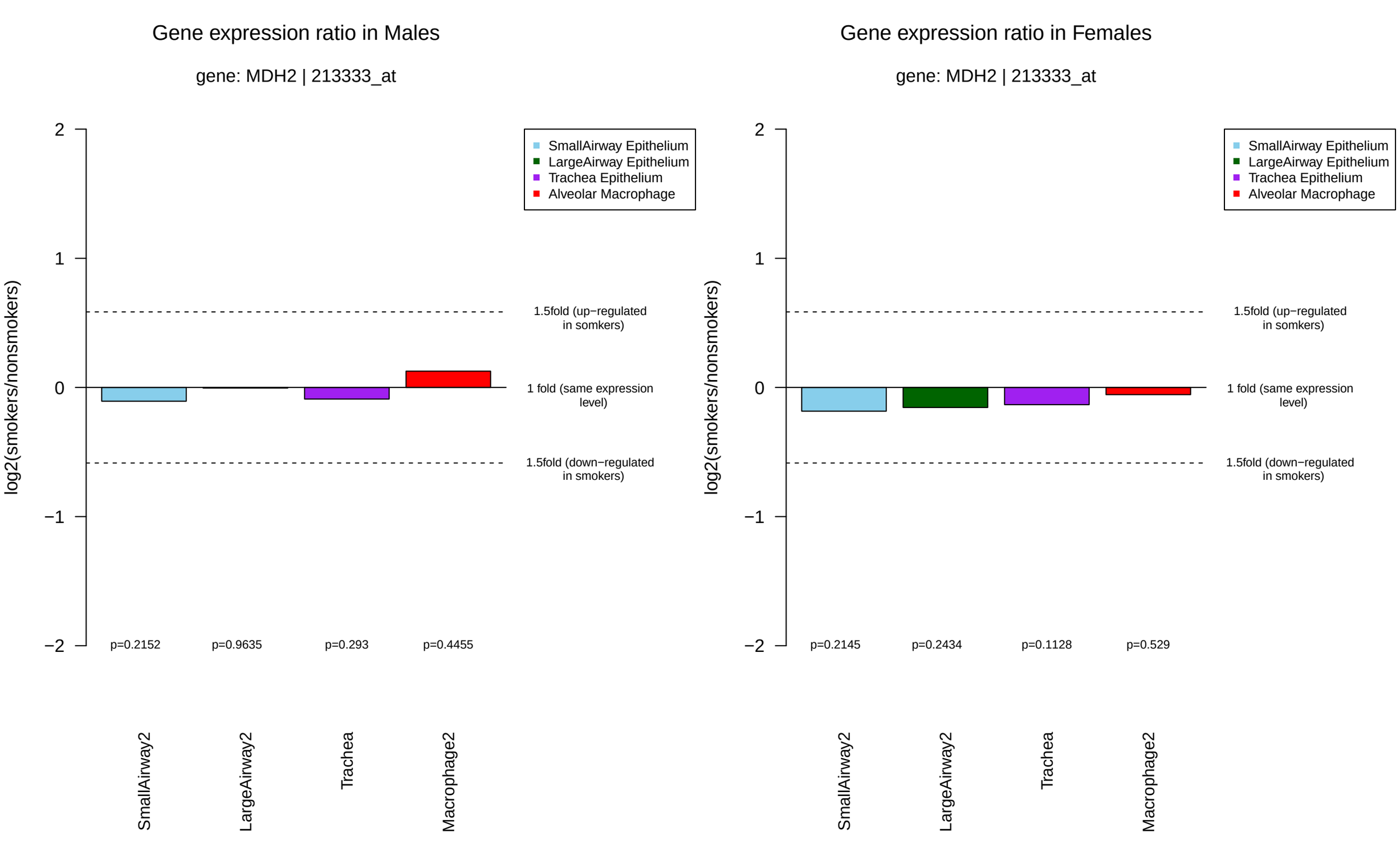
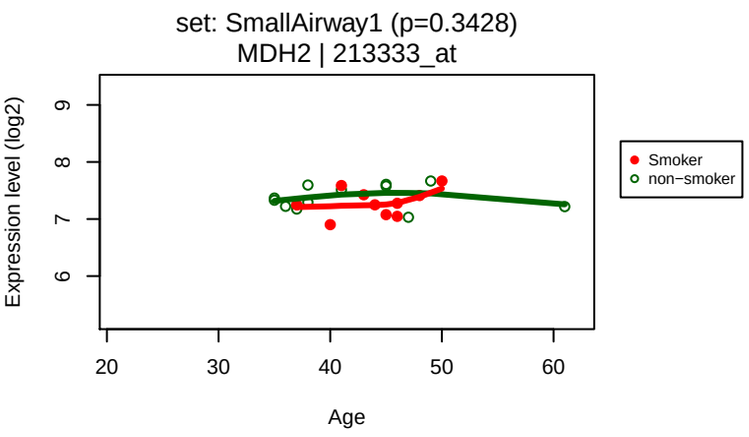
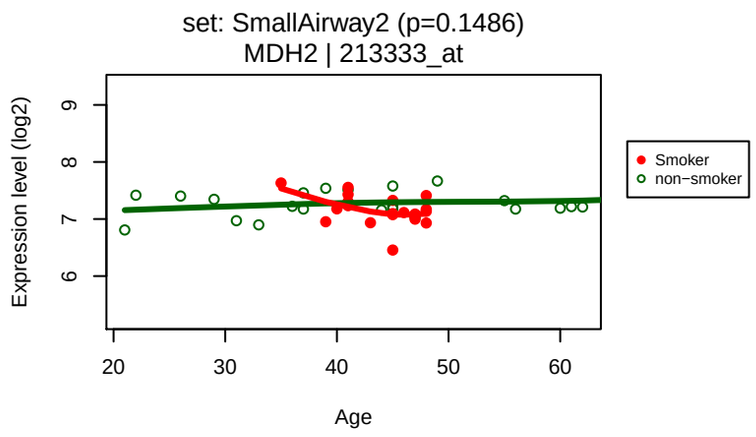
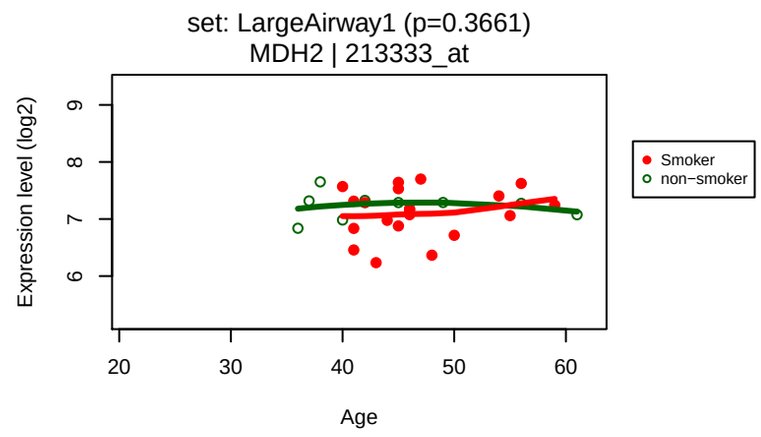

Gene expression of smokers and nonsmokers
gene: MDH2 | 213333_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3428
p=0.1486
p=0.1517
p=0.3661
p=0.4469
p=0.132
p=0.8297
p=0.5204
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

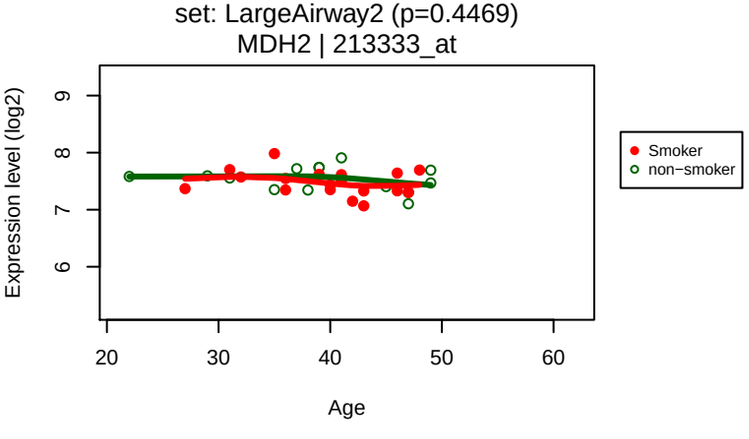

Gene expression ratio (smokers/nonsmokers)
gene: MDH2 | 213333_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3428
p=0.1486
p=0.1517
p=0.3661
p=0.4469
p=0.132
p=0.8297
p=0.5204
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

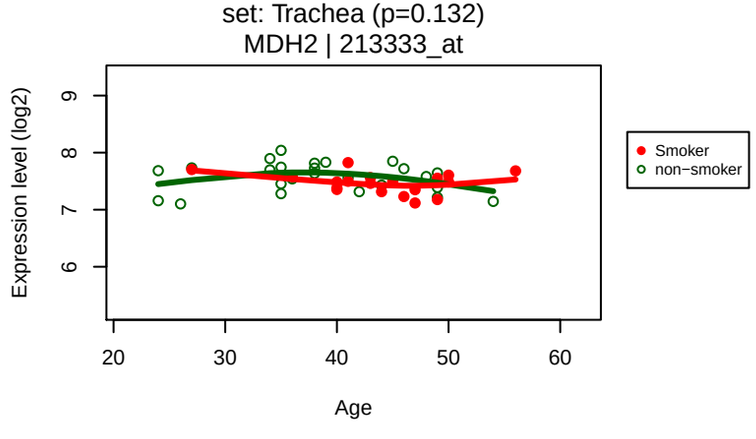

Gene expression ratio in Males
gene: MDH2 | 213333_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2152
p=0.9635
p=0.293
p=0.4455
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MDH2 | 213333_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2145
p=0.2434
p=0.1128
p=0.529
SmallAirway2
LargeAirway2
Trachea
Macrophage2

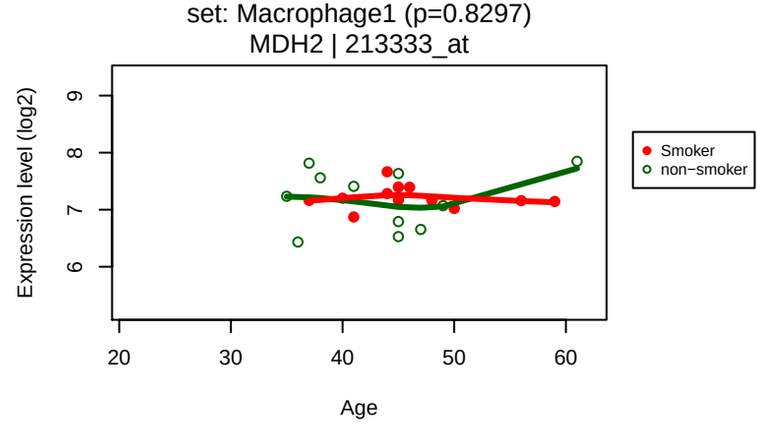

set: SmallAirway1 (p=0.3428)
MDH2 | 213333_at
Expression level (log2)
Age
Smoker
non−smoker

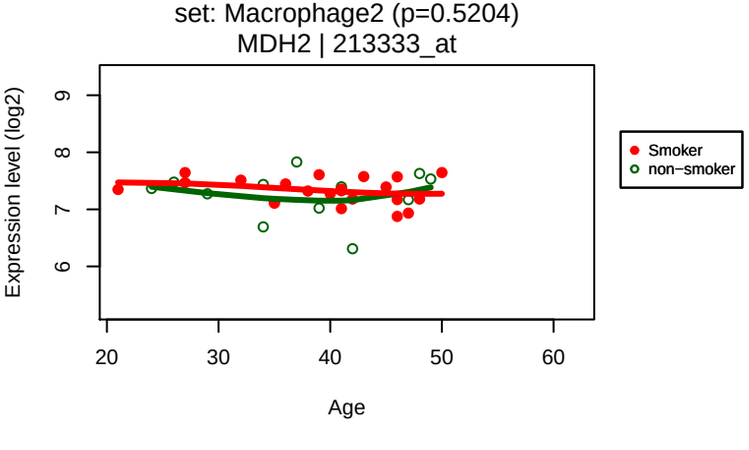

set: SmallAirway2 (p=0.1486)
MDH2 | 213333_at
Expression level (log2)
Age
Smoker
non−smoker

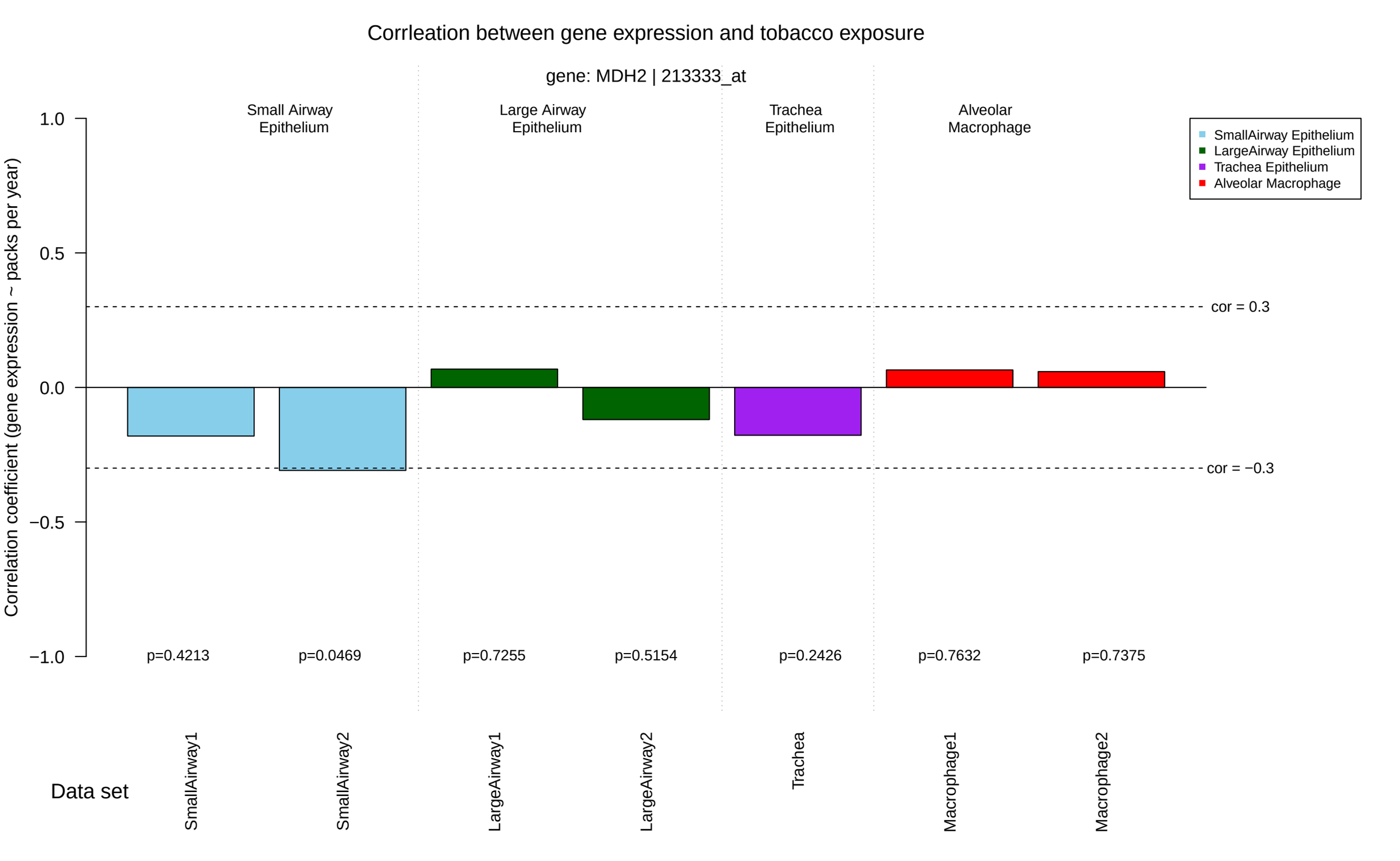

set: LargeAirway1 (p=0.3661)
MDH2 | 213333_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4469)
MDH2 | 213333_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.132)
MDH2 | 213333_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8297)
MDH2 | 213333_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5204)
MDH2 | 213333_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MDH2 | 213333_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4213
p=0.0469
p=0.7255
p=0.5154
p=0.2426
p=0.7632
p=0.7375
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2